

Configuring SSH Remote Access in Packet Tracer

Standard Operating Procedure & Implementation Guide for Cisco Routers (/28 Subnet)

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Environment: Cisco Packet Tracer

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1. Network Topology & Overview

This documentation outlines the step-by-step procedure to enable secure remote management (SSH v2) on a Cisco Router. The network utilizes a /28 prefix (Subnet Mask: 255.255.255.240). The PC connects to the router through a Layer 2 Switch and initiates remote management via the Packet Tracer Desktop SSH/Telnet GUI tool.

[PC0: 192.168.1.3/28] → [Switch (SW1): 192.168.1.2/28] → [Router (Main): 192.168.1.1/28]

2. IP Addressing Scheme (/28 Subnet)

DEVICE NAME	INTERFACE	IP ADDRESS	SUBNET MASK	DEFAULT GATEWAY	ROLE
Router (Main)	GigabitEthernet 0/0/0	192.168.1.1	255.255.255.240 (/28)	N/A	Default Gateway & SSH Server
Switch (SW1)	VLAN 1 (SVI)	192.168.1.2	255.255.255.240 (/28)	192.168.1.1	Layer 2 Switch (Management SVI)
PC0	FastEthernet 0	192.168.1.3	255.255.255.240 (/28)	192.168.1.1	SSH Client Workstation

3. Router Configuration Steps (Cisco IOS)

Step 1 Configure Default Gateway LAN Interface with /28 Subnet Mask

Assign IP 192.168.1.1 with subnet mask 255.255.255.240 to the GigabitEthernet interface.

```
Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet 0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.240
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 2 Define Hostname (Main) and Domain Name

Set the hostname to **Main** and specify the domain name required for RSA key generation.

```
Router(config)# hostname Main
Main(config)# ip domain-name lab.local
```

Step 3 Generate RSA Cryptographic Key Pair

Generate the RSA key pair required for SSH encryption using a standard 2048-bit modulus.

```
Main(config)# crypto key generate rsa
How many bits in the modulus [512]: 2048
```

Step 4 Create Local Admin Account & Enforce SSHv2

Define an administrator account with privilege level 15 and restrict protocol operation to SSH Version 2.

```
! Configure local administrator user
Main(config)# username admin privilege 15 secret RouterPass123!

! Enforce SSH Version 2
Main(config)# ip ssh version 2
```

Step 5 Configure Virtual Terminal (VTY) Lines

Restrict incoming remote access on virtual lines 0 through 15 to SSH only.

```
Main(config)# line vty 0 15
Main(config-line)# transport input ssh
Main(config-line)# login local
Main(config-line)# exit
```

4. Testing Remote Access via Packet Tracer Desktop SSH App

A Connecting via Desktop TELNET/SSH Client GUI

1. Click on **PC0** (IP: 192.168.1.3, Mask: 255.255.255.240) in Packet Tracer and navigate to the **Desktop** tab.
2. Open the **TELNET/SSH Client** application.
3. Select **SSH** as the radio button option.
4. Enter Host IP: 192.168.1.1 and Username: admin.
5. Click **Connect** and enter password RouterPass123! when prompted.

B Verify Active Session on Router

Run operational commands on router **Main** to confirm active SSH sessions:

```
Main# show ip ssh
Main# show ssh
```

Subnet Architecture Note (/28)

With a **/28 prefix** (255.255.255.240), the subnet provides 14 usable host IP addresses ranging from 192.168.1.1 to 192.168.1.14. All three devices (Router .1, Switch .2, PC0 .3) reside within the same local broadcast domain.